

First/contact author & main contributor

- [1] A. A. Vigliano and F. Longo. *Soccer Ball Tessellation: a novel method for tiling large uncertainty areas with narrow field of view telescopes*. In preparation. 2026.
Article based on the results of the PhD thesis and its new developments
- [2] A. A. Vigliano, F. Longo, and Ž. Bošnjak. “Reassessing high-energy emission correlations in gamma-ray bursts using a large, homogeneous sample of X-ray afterglows”. In: (May 2026). A&A, accepted. DOI: [10.1051/0004-6361/202558494](https://doi.org/10.1051/0004-6361/202558494). arXiv: [2605.25644](https://arxiv.org/abs/2605.25644) [astro-ph.HE].
Conceived the analysis, developed the methodology, performed the data analysis, wrote the paper
- [3] A. A. Vigliano. “Novel strategies for the observations of fast gamma-ray transients”. Publicly available [online](#). PhD thesis. Università degli Studi di Udine, Udine, Italy, 2025.
PhD thesis. I conceived the analysis, developed the methodology, performed the data analysis, wrote the paper
- [4] A. A. Vigliano and F. Longo. “Gamma-ray Bursts: 50 Years and Counting!” In: *Universe* 10.2 (2024), p. 57. DOI: [10.3390/universe10020057](https://doi.org/10.3390/universe10020057).
A review of the history of gamma-ray burst observations and studies. I conducted in-depth historical and scientific research and wrote the manuscript.
- [5] A. Kouchner, A. A. Vigliano, and M. Lamoureux. “Detection and phenomenology of cosmic neutrinos”. In: *PoS QG-MMSchools* (2024), p. 005. DOI: [10.22323/1.440.0005](https://doi.org/10.22323/1.440.0005).
A review of the history of observations and studies of astrophysical neutrinos as part of the Second School of the COST Action QGMM. I conducted extensive historical and scientific research and wrote the manuscript.
- [6] A. A. Vigliano et al. “Gamma Ray Burst localization with GECCO’s BGO anti-coincidence shields”. In: *PoS ICRC2023* (2023), p. 748. DOI: [10.22323/1.444.0748](https://doi.org/10.22323/1.444.0748).
Conceived the analysis, developed the methodology, performed the data analysis, wrote the paper

Contributor

- [7] R. Aloisio et al. “Crystal Eye: All sky MeV monitor with high precision real-time localization”. In: *Astroparticle Physics* 174 (2026), p. 103171. ISSN: 0927-6505. DOI: <https://doi.org/10.1016/j.astropartphys.2025.103171>. URL: <https://www.sciencedirect.com/science/article/pii/S0927650525000945>.
Contributed to the evaluation of the localization capabilities of transient events (especially GRBs) through classical and ML techniques
- [8] R. Alves Batista et al. “White paper and roadmap for quantum gravity phenomenology in the multi-messenger era”. In: *Class. Quant. Grav.* 42.3 (2025), p. 032001. DOI: [10.1088/1361-6382/ad605a](https://doi.org/10.1088/1361-6382/ad605a). arXiv: [2312.00409](https://arxiv.org/abs/2312.00409) [gr-qc].
Contributed to writing and review. Active member of the COST Action QGMM and co-proposer of its followup COST Action BridgeQG
- [9] A. Campoy-Ordaz et al. “Search for Lorentz Invariance Violation with time-lag on gamma-ray Cherenkov Telescope data : From the data to the time lag constraints”. In: *PoS QG-MMSchools* (2024), p. 009. DOI: [10.22323/1.440.0009](https://doi.org/10.22323/1.440.0009).
Contributed to the writing (tutorial section of the paper)
- [10] A. Coogan et al. “Hunting for dark matter and new physics with GECCO”. In: *Phys. Rev. D* 107.2 (2023), p. 023022. DOI: [10.1103/PhysRevD.107.023022](https://doi.org/10.1103/PhysRevD.107.023022). arXiv: [2101.10370](https://arxiv.org/abs/2101.10370) [astro-ph.HE].
Contributed with the evaluation of the detector’s capabilities
- [11] A. Donini et al. “Performance study update of observations in divergent mode for the Cherenkov Telescope Array”. In: *PoS ICRC2023* (2023), p. 840. DOI: [10.22323/1.444.0840](https://doi.org/10.22323/1.444.0840). arXiv: [2309.14106](https://arxiv.org/abs/2309.14106) [astro-ph.HE].
Contributed as an expert on tessellation strategies for the follow-up of poorly localized transient events

The publications listed below are large-scale collaborative works. I am member of both analysis and operations teams contributing to data taking, validation, and internal reviews.

Collaboration papers

- [12] K. Abe et al. “Long-term study of the gamma-ray emission of Cygnus X-3 with MAGIC and Fermi-LAT”. In: (Apr. 2026). arXiv: [2604.27671 \[astro-ph.HE\]](#).
- [13] K. Abe et al. “The January 2010 flare of Mrk421: Insights from a stochastic acceleration model”. In: (Apr. 2026). arXiv: [2604.20480 \[astro-ph.HE\]](#).
- [14] K. Abe et al. “Extreme Blazars Observed with MAGIC: Second Catalog Release”. In: *Astrophys. J.* 1002.2 (2026), p. 138. DOI: [10.3847/1538-4357/ae5a8a](#). arXiv: [2604.05576 \[astro-ph.HE\]](#).
- [15] S. Abe et al. “MAGIC observations of NGC 4278. The first low-luminosity radio galaxy with compact jets detected at TeV energies”. In: (Mar. 2026). arXiv: [2603.22955 \[astro-ph.HE\]](#).
- [16] K. Abe et al. “VHE gamma-ray intranight variability from BL Lacertae during the extreme flaring state of 2022”. In: (Mar. 2026). arXiv: [2603.19889 \[astro-ph.HE\]](#).
- [17] K. Abe et al. “First detection of VHE gamma-ray signal from the FSRQ TON 0599”. In: *Mon. Not. Roy. Astron. Soc.* 546 (2026), p. 4. DOI: [10.1093/mnras/stag032](#). arXiv: [2601.03873 \[astro-ph.HE\]](#).
- [18] J. Abhir et al. “Prompt Searches for Very-high-energy γ -Ray Counterparts to IceCube Astrophysical Neutrino Alerts”. In: *Astrophys. J.* 997.2 (2026), p. 141. DOI: [10.3847/1538-4357/ae2c4e](#). arXiv: [2512.16562 \[astro-ph.HE\]](#).
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- [21] M. Tambone et al. “WINK: Advancing X and Gamma Ray Detection Technology for Space Observations”. In: *PoS ICRC2025* (2025), p. 857. DOI: [10.22323/1.501.0857](#).
- [22] A. Tarana et al. “X-ray Binaries: A Key Corollary Science with Crystal Eye”. In: *PoS ICRC2025* (2025), p. 594. DOI: [10.22323/1.501.0594](#).
- [23] K. Abe et al. “Time-dependent Modeling of the Subhour Spectral Evolution during the 2013 Outburst of Mrk 421”. In: *Astrophys. J.* 998.1 (2026), p. 6. DOI: [10.3847/1538-4357/ae25f0](#). arXiv: [2509.08686 \[astro-ph.HE\]](#).
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